Short-term PM2.5 Forecasting on the Beijing Multi-Site Air Quality Dataset

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Why forecast PM2.5?

- PM2.5 = fine particulate matter, the main driver of haze in dense cities; linked to respiratory and cardiovascular risk.
- $\geq 75\,\mu g/m^3$ is the WHO / Chinese GB "unhealthy" threshold.
- Short-horizon forecasts let cities issue advisories *before* a spike hits.

Two questions we set out to answer:

1. Do sequence models (LSTM / GRU / Transformer) beat feature-based regressors here?
2. Does context from the *other* 11 stations help?

Data
Beijing Multi-Site Air Quality (UCI):

- 12 monitoring stations
- hourly readings, 2013-03 – 2017-02
- 6 pollutants + 6 weather variables
- $\sim$35 000 hours per station

Target
PM2.5 at $t+h$, $h \in \{1, 6\}$ hours ahead.

Setup: windows, splits, two input settings

Sliding window
Input: last 24 hours of (pollutants, weather, time, station).
Target: PM2.5 at $t+h$.

Time-based split (no leakage)

Train	2013-03 – 2015-12
Val	2016-01 – 2016-06
Test	2016-07 – 2017-02

Preprocessing
Standardize on train only; clip to $\pm 8\sigma$ (rainfall outliers were causing GRU NaN losses).
Cyclic $(\sin, \cos)$ encoding for wind direction, hour-of-day, day-of-week, month-of-year.

Two input settings

- global – target station's own 24 h history; station id as embedding (DL) / one-hot (baselines).
  $F = 19$ features per timestep.
- multi-site – same as above, but every timestep also gets the channel-wise *mean of the other 11 stations*. Target station masked out (no leakage).
  $F = 38$ features per timestep.

Metrics
MAE, RMSE, R^2, and spike-recall = fraction of test hours with $y \geq 75\,\mu g/m^3$ where the model also predicts ≥ 75 (i.e. would we have flagged it?).

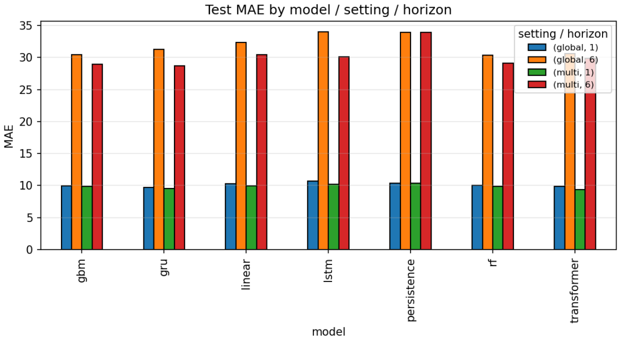
Seven models, one pipeline

Family	Models	Notes
no learning	**Persistence** ($\hat{y}_{t+h} = y_t$)	Tough baseline at h=1; collapses at h=6.
feature-based	**Ridge**, **Random Forest**, **Gradient Boosting**	Flatten $24 \times F$ window; append one-hot station id.
sequence	**LSTM**, **GRU**, **Transformer enc.**	2 layers, hidden 128, dropout 0.2; 8-dim station embedding.

Training (sequence models). AdamW (lr 10^{-3}, weight decay 10^{-5}), cosine LR, batch 512, grad-clip 1.0, smooth-L_1 loss in standardized space, max 15 epochs, early stop on val MAE (patience 4). PyTorch MPS backend (Apple Silicon GPU).

Same data, same targets, same metrics across all seven models $\Rightarrow$ apples-to-apples comparison of model families and of the two input settings.

Result 1: GRU multi-site is the best at both horizons



Test MAE per model / input setting / horizon (lower is better).

1 hour ahead

Persistence already gets $R^2 \approx 0.95$. **Transformer multi-site MAE = 9.41**, **GRU multi-site = 9.51** - both edge past every baseline. Spike-recall $\approx$ **0.95** ($\geq 75\,\mu g/m^3$).

6 hours ahead

Persistence collapses ($R^2 \approx 0.56$). **GRU multi-site MAE = 28.75** edges out the best baseline (GBM multi-site, 29.00). LSTM struggles at our default LR / patience.

Take-away: sequence models help most when paired with the multi-site context.

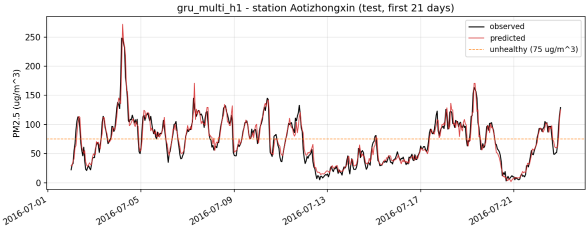
Result 2: a trivial spatial summary already helps

Multi-site setting: at every timestep feed the model the channel-wise *mean of the other 11 stations*.
Doubles F from 19 to 38.
No graph, no attention – the simplest possible spatial summary.

Test-MAE change vs. global ($\mu g/m^3$):

Model	Δ at h=1	Δ at h=6
Ridge	-0.38	-1.87
Random Forest	-0.18	-1.26
GBM	-0.11	-1.43
GRU	-0.19	$-$**2.55**
Transformer	-0.43	-0.79

Consistent across all models; *larger* gains at h=6 where local history alone leaves more to predict. A wind-aware aggregator should help more still.



GRU (multi-site, h=1) at Aotizhongxin station, first three weeks of test. Tracks the daily cycle and the multi-day haze episodes; dashed orange = 75 $\mu g/m^3$ unhealthy threshold.

Wrap-up

What we found

- At h=1, GRU / Transformer (multi-site) reach MAE $\approx$ 9.4 and recall $\sim$95% of unhealthy hours.
- At h=6, GRU (multi-site) is the best single model (MAE 28.75), narrowly above GBM (29.00).
- A trivial *spatial mean* of the other 11 stations gives a consistent MAE boost - larger at h=6 (-1 to $-4\,\mu g/m^3$).

Next steps

- Wind-direction-aware spatial aggregation (upwind stations matter more).
- Proper LR / patience sweep tuned for h=6.
- Multi-task forecasting (PM10, NO_2, ...) for a shared representation.

Reproducible. Code, training, evaluation, figures, and the full report all live in the project repository.

Thanks – questions?